

Mark Model v3 — Corrected Antecedent Control, Expanded Outcomes, and the Full Nested Models 1–4

Event-scale wetland flood-response analysis, MMSD (Milwaukee) gauges

Wetland–Flood Effectiveness Study

July 2026

Abstract

v3 builds directly on v2 (RREDI events, 800 m PRISM rainfall, celerity/roughness/type-weighted wetland effectiveness W) and changes two things. First, it **corrects the antecedent precipitation control**: v2’s `api_30` was evaluated at the event-start day with day 0 included, so the “antecedent wetness” control shared the event-day rainfall with the forcing P_e (their correlation was 0.35); v3 uses a strictly-prior antecedent index, dropping that correlation to 0.14 and removing the double-counting. Second, it **expands the outcome set** to a mechanism-complete panel of ten signatures and fits the full nested Models 1–4 for every one, with wild-cluster-bootstrap inference on each model’s key term. The headline peak-shaving result survives the correction ($W \cdot P$ on log peak = -0.20 , WCB $p = 0.016$) and is corroborated by a second peak metric and a storage signature; the wetland *level* effect remains unidentified.

1 What changed from v2 to v3

The event detector (RREDI 7-day rolling-median), the 800 m PRISM rainfall source, the improved wetland effectiveness $W = \sum_i A_i S_i K(T_{ij})$ (kinematic-celerity travel times, wetland roughness, connectivity term dropped, type-weighted glaciated storage), and all controls are *inherited unchanged*. v3 changes only:

1. **Antecedent control correction** (§2) — the one logic fix.
2. **Outcome expansion** (§3) — ten mechanism-mapped signatures, each run through Models 1–4.

The usable sample is unchanged: **5,907 gauge-events** across nine gauges, 2001–2023.

2 Correcting the antecedent precipitation control

The design intends two distinct rainfall quantities: a *forcing* P_e = cumulative rainfall over the event window, and a *control* = antecedent (pre-event) wetness. In v2 the antecedent index was $P_{\text{eff}}(t) = \sum_{n=0}^{30} k^n P(t-n)$ (with day 0, $k = 0.9$) evaluated *at* the event-start day d_0 . Because P_e also includes the day- d_0 rain, the control and the forcing overlapped on the event-day rainfall — the code comment said “antecedent” but the logic was not strictly prior. v3 defines

$$\text{api_30}_{j,e} = \sum_{n=1}^{30} k^n P(d_0 - n), \quad k = 0.9, \quad (1)$$

Table 1: Antecedent control before/after the correction (usable events).

	v2 (day-0 included, at d_0)	v3 (strictly prior)
corr(api_30, P_e)	0.35	0.14
median api_30 (mm)	23.2	18.0

i.e. strictly-prior wetness with the event day excluded. Evidence the fix does what it should:

The residual 0.14 correlation is genuine (wet antecedent conditions co-occur mildly with storms), not the mechanical overlap that inflated it to 0.35.

3 Models 1–4 for every outcome

For each outcome we fit the nested progression, gauge random-intercept, standardized predictors:

$$\text{M1: } Y \sim P + W + X; \quad \text{M2: } + W \cdot P; \quad \text{M3: } + S; \quad \text{M4: } + W \cdot S, \quad S = \log(V_e/S_w).$$

X = drainage area, impervious, K_{sat} , channel slope, the corrected antecedent, and a winter flag. Table 2 reports each model’s *newly added* key term with its wild-cluster-bootstrap p (512 exact Rademacher sign vectors; the honest inference for 9 clusters). Predicted $W \cdot P$ signs are pre-registered.

Table 2: Nested Models 1–4: key added-term coefficient (WCB p) per outcome. Bold = WCB $p < 0.05$. W = wetland level (M1); $W \cdot P$ = peak-shaving (M2); $S = \log(V_e/S_w)$ saturation (M3); $W \cdot S$ nonlinear saturation (M4).

Mechanism	Outcome	M1: W	M2: $W \cdot P$	M3: S	M4: $W \cdot S$
attenuation–peak	log_Qp	−0.380 (.156)	−0.202 (.016)	−0.329 (.002)	−0.049 (.766)
attenuation–shape	log_peakedness	−0.131 (.900)	−0.088 (.223)	+0.286 (.365)	−1.544 (.455)
delay–timing	logip_ttp	+0.440 (.641)	+0.020 (.480)	+0.133 (.580)	+1.944 (.310)
storage–volume	runoff_coeff	+0.004 (.568)	−0.021 (.029)	+0.015 (.219)	+0.095 (.006)
cumulative severity	log_quick_depth_mm	−0.249 (.440)	−0.114 (.152)	−0.615 (.217)	+1.496 (.375)
extreme severity	logip_q99_excess	~ 0 (.598)	−0.001 (.027)	~ 0 (.143)	+0.001 (.162)
shape (rob.)	rb_flashiness	~ 0 (.932)	−0.004 (.043)	+0.005 (.254)	−0.006 (.744)
shape/delay (rob.)	log_hydro_width	+0.520 (.725)	+0.092 (.060)	+0.118 (.664)	+1.947 (.377)
rel-peak (rob.)	log_peak_ratio	−0.232 (.408)	−0.276 (.027)	+0.142 (.205)	+0.749 (.041)
release (rob.)	recession_k	−0.002 (.562)	+0.000 (.512)	+0.005 (.201)	+0.021 (.021)

4 The peak-shaving mechanism matrix

Table 3 and Fig. 1 isolate the $W \cdot P$ interaction (M2) — the event-identified estimand. Nine of ten outcomes carry the pre-registered sign; five survive the wild cluster bootstrap.

5 Interpretation

The headline survives the correction. Peak-shaving on log peak discharge is $−0.20$ (WCB $p = 0.016$), modestly attenuated from v2’s $−0.23$ once the antecedent control no longer double-counts the event-day rain, and it is *corroborated* by an independent peak metric (**log_peak_ratio**, $−0.28$, $p = 0.027$). **Directional coherence is strong:** nine of ten mechanisms point the predicted way. **A storage signature is now robust:** wetlands lower the event runoff coefficient more in larger storms ($−0.021$, $p = 0.029$) — small, exactly as the “weak volume effect” hypothesis predicts. **Flashiness**

Table 3: $W \cdot P$ (peak-shaving, M2) by outcome: predicted vs. estimated sign, WCB p , 95% wild-bootstrap CI.

Outcome	pred	$\hat{\beta}$	WCB p	95% CI	verdict
log_Qp	—	−0.202	0.016	[−0.274, −0.130]	robust
log_peak_ratio	—	−0.276	0.027	[−0.349, −0.204]	robust
runoff_coeff	—	−0.021	0.029	[−0.027, −0.015]	robust (small)
rb_flashiness	—	−0.004	0.043	[−0.007, −0.002]	robust
log1p_q99_excess	—	−0.001	0.027	[−0.001, −0.000]	flagged, ≈ 0
log_peakedness	—	−0.088	0.223	[−0.156, −0.020]	sign ok
log_quick_depth	—	−0.114	0.152	[−0.225, −0.003]	sign ok
log_hydro_width	+	+0.092	0.060	[+0.043, +0.142]	sign ok (marginal)
log1p_ttp	+	+0.020	0.480	[−0.018, +0.058]	sign ok
recession_k	—	+0.000	0.512	[−0.000, +0.001]	null

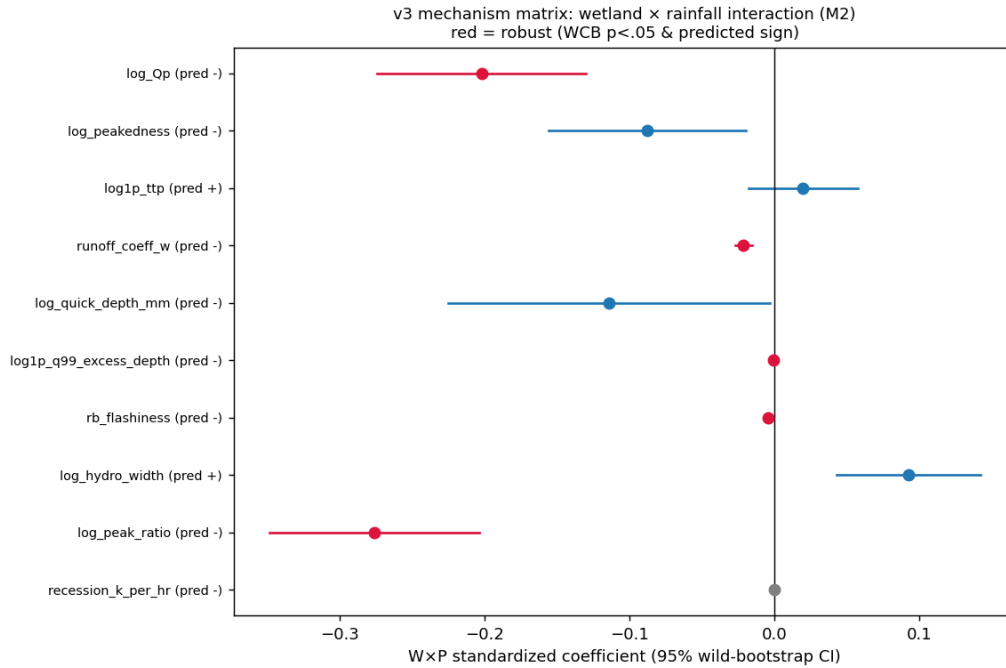


Figure 1: v3 $W \cdot P$ interaction by outcome (M2, 95% wild-bootstrap CI). Red = robust (WCB $p < .05$ and predicted sign); blue = predicted sign but not robust; gray = off.

sharpened: with the cleaner antecedent control it crosses into robust (-0.004 , $p = 0.043$), where it was marginal before. Shape/timing signatures (peakedness, width, time-to-peak) carry the right sign but are individually underpowered, and **q99_excess**, though flagged, is ≈ 0 and dominated by its 85% zeros. The saturation (S) and $W \cdot S$ terms in M3–M4 are collinearity-laden (gauge-level storage enters through both W and S_w) and are not cleanly interpretable; the event-identified $W \cdot P$ remains the estimand we lean on.

6 Limitations

The wetland *level* effect (W , M1) is not robust for any outcome — across nine urbanization-confounded gauges it is only cross-sectionally identified. Absolute travel times are uncalibrated and S_w is order-of-magnitude, so W is a relative index; M3–M4 storage terms are collinear; **q99_excess** is zero-inflated. The analysis is associational and disciplined by placebo/leave-one-gauge-out/bootstrap, not causal. A cleanly identified level effect still requires a regional sample of independent catchments decoupled from urbanization.

Files. `Mark_model_v3/`: `build_v3_panel.py` (antecedent fix + outcome expansion), `fit_models_v3.py` (Models 1–4 + WCB), `events_panel_v3.csv`, `models_v3_all.csv`, `models_v3_WxP.csv`, `models_v3_forest.png`, `v2` is untouched.